PRECOMPOSITION ON CONTINUOUS ALTERNATING MAPS IS NOT ANALYTIC IN CHARACTERISTIC 2

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Abstract. For normed spaces E, E', F over a nontrivially normed field k and a finite ι, precomposition $f \mapsto (m \mapsto m \circ (f, \dots, f))$ sends continuous linear maps $E \to E'$ to continuous linear maps between spaces of continuous alternating maps. It is C^n for every $n \in \mathbb{N} \cup \{\infty\}$ in every characteristic [5], and analytic when $|\iota|!$ is invertible in k (by polarisation). We show that it is *not* analytic in general: over every complete ultrametric field of characteristic 2 that is not spherically complete, with $E = E' = \ell^\infty$ and $|\iota| = 2$, analyticity fails at 0. The proof is a finite parity identity for determinants in characteristic 2, glued along ℓ^∞; the failure is witnessed by two coordinate projections of ℓ^∞. The result is formalized in Lean 4/Mathlib in the form "the general C^ω statement implies `False`".

1. Introduction

Let k be a nontrivially normed field. For normed spaces E, E', F over k and a finite ι we write $\mathcal{L}(E, E')$ for the space of continuous linear maps with the operator norm, and $\mathrm{Alt}^\iota(E, F)$ for the normed space of continuous alternating ι-linear maps $E^\iota \to F$; we write Alt^2 when $|\iota| = 2$.

Analytic functors. A map $f\colon X \to Z$ between normed spaces over k is **analytic at** $x_0 \in X$ if there are bounded n-linear maps $p_n\colon X^n \to Z$ and a radius $r > 0$ with $\sup_n \|p_n\| r^n < \infty$ such that $f(x_0 + h) = \sum_{n \geq 0} p_n(h, \dots, h)$ whenever $\|h\| < r$. It is **analytic**, or C^ω, if it is analytic at every point of X; this is the definition used in Mathlib [4].

Lang [3, Ch. III, §4] calls a functor T on Banach spaces **of class C^p** if $f \mapsto T(f)$ is C^p. Similarly, we call a contravariant endofunctor T on the category of normed spaces over k **analytic** if for all E, E' the map

$$\mathcal{L}(E, E') \to \mathcal{L}\big(T(E'), T(E)\big), \qquad f \mapsto T(f),$$

is analytic. For a normed space F and a finite ι, the assignment $E \mapsto \mathrm{Alt}^\iota(E, F)$, $f \mapsto (m \mapsto m \circ (f, \dots, f))$, is a contravariant functor. The assignment is also functorial in F: a continuous linear map $g\colon F \to F'$ acts by $m \mapsto g \circ m$. This action is linear in g, hence analytic, whereas the action of f is not linear in f. So the bifunctor Alt^ι is analytic if and only if the functor $\mathrm{Alt}^\iota(-, F)$ is analytic for every F.

Motivation: alternating bundles in Mathlib. In Mathlib, manifolds and vector bundles are modelled on normed spaces (of possibly infinite dimension) over an arbitrary nontrivially normed field k, and their regularity is an element $n \in \mathbb{N} \cup \{\infty, \omega\}$, where C^ω means analytic. Differential forms require the bundle of continuous alternating maps between two C^n vector bundles to be a C^n vector bundle again (an instance of `ContMDiffVectorBundle`). Its transition functions are those of the given bundles with Alt^ι applied, so this comes down to the regularity of the functor $\mathrm{Alt}^\iota(-, F)$, i.e. of the precomposition map

`ContinuousAlternatingMap.compContinuousLinearMapCLM`.

Two cases have been settled, in pull requests to Mathlib.

- If $|\iota|!$ is invertible in k then $\mathrm{Alt}^\iota(-,F)$ is analytic, by polarisation. In Mathlib this is done in characteristic 0 by Gouëzel [1], who uses it to construct the bundle for every n in characteristic 0 [2]; he remarks that a variant of the argument, not included there, covers finite-dimensional spaces in positive characteristic, but not the general case.
- Over every k, with no hypothesis on the characteristic or the dimension, $\mathrm{Alt}^\iota(-,F)$ is of class C^n for every $n \in \mathbb{N} \cup \{\infty\}$ [5].

This leaves open the case where $|\iota|!$ is not invertible in k and the spaces are infinite-dimensional: is $\mathrm{Alt}^\iota(-,F)$ analytic there? The following theorem provides a counterexample.

Theorem 1. *Let k be a complete ultrametric field of characteristic 2 which is not spherically complete. There is a Banach space F over k such that the functor $\mathrm{Alt}^2(-,F)$ is not analytic.*

Such fields exist (Proposition 12), so the restriction $n \neq \omega$ in [5] cannot be removed. In characteristic $\ell \geq 3$ the case $|\iota| \geq \ell$ remains open (Remark 15). The result is machine-checked in Lean 4 with Mathlib (§9).

2. Statement

Notation. For a polynomial f and a monomial μ, $\operatorname{coeff}_\mu f$ is the coefficient of μ in f. We write $c_0 \subset \ell^\infty$ for the null sequences, $\mathbf{1}_S$ for the indicator of a set S, and $\mathbf{1} = \mathbf{1}_{\mathbb{N}}$.

Throughout, k is a field which is complete for a nontrivial absolute value $|\cdot|$ satisfying the **ultrametric inequality** $|x+y| \leq \max(|x|,|y|)$; we call such a field **ultrametric**. (Absolute values with this property are also called non-archimedean; this is a condition on the absolute value and is unrelated to non-Archimedean *ordered* fields.) The sup norms used below inherit the inequality. We assume that

(K1) $\operatorname{char} k = 2$;
(K2) k is not spherically complete: there is a sequence of closed balls $B(c_n, r_n)$ such that $|c_{n+1} - c_n| \leq r_n$, $r_0 > r_1 > \cdots$ and $\bigcap_n B(c_n, r_n) = \emptyset$.

The balls in (K2) are balls in k, and they are nested by the ultrametric inequality. Such fields exist; an explicit one is given in §7. Let $E = \ell^\infty = \ell^\infty(\mathbb{N}, k)$ with the sup norm, a Banach algebra under pointwise multiplication; write $m_a \in \mathcal{L}(E,E)$ for multiplication by $a \in E$, so that $\|m_a\| = \|a\|$ (evaluate at $\mathbf{1}$), and e_i for the unit vectors. For $x, y \in E$ let

$$W(x,y) := x \otimes y + y \otimes x \in \ell^\infty(\mathbb{N} \times \mathbb{N}, k), \qquad W(x,y)_{pq} = x_p y_q + x_q y_p,$$

and let $F \subseteq \ell^\infty(\mathbb{N} \times \mathbb{N}, k)$ be the closed linear span of all $W(x,y)$, a Banach space. Let

$$Q \colon \mathcal{L}(E,E) \to \mathcal{L}\big(\mathrm{Alt}^2(E,F), \mathrm{Alt}^2(E,F)\big), \qquad Q(f)(m) = m \circ (f,f).$$

Theorem 2. *Q is not analytic at 0. In particular Q is not C^ω.*

The map Q is the action on $\mathcal{L}(E,E)$ of the functor $\mathrm{Alt}^2(-,F)$, so Theorem 2 implies Theorem 1. Sections 3–6 prove Theorem 2 in four steps.

3. Step 1: analyticity forces a bilinear lift

Lemma 3. *Let $Q \colon X \to Z$ be a map between normed spaces with $Q(tf) = t^2 Q(f)$. If Q is analytic at 0 then there is a bounded bilinear $P \colon X \times X \to Z$ with $P(f,f) = Q(f)$ for all f.*

Proof. Let $\sum_n p_n$ be a power series of Q at 0. For fixed f the one-variable power series $\sum_n t^n\, p_n(f,\dots,f)$ and the polynomial $t^2 Q(f)$ agree for small t. Power series coefficients in one variable are unique over any nontrivially normed field, so $p_2(f,f) = Q(f)$. Take $P = p_2$. (No division by 2 occurs, and P need not be symmetric.) □

So assume that a bounded bilinear P with $P(f,f) = Q(f)$ exists. The map $W\colon E \times E \to F$ is bilinear, it is alternating because $W(x,x) = 2\,x \otimes x = 0$, and $\|W(x,y)\| \le \|x\|\|y\|$ by the ultrametric inequality; so $W \in \mathrm{Alt}^2(E,F)$ with $\|W\| \le 1$. Put, for $a, b, x, y \in E$,

$$\Phi(a,b;x,y) := P(m_a, m_b)(W)(x,y) \in F.$$

Then Φ is 4-linear, and since $\|m_a\| = \|a\|$ and $\|W\| \le 1$ we have

$$\|\Phi(a,b;x,y)\| \le C\|a\|\,\|b\|\,\|x\|\,\|y\|, \qquad C := \max(1, \|P\|).$$

Moreover

$$\Phi(a,b;x,x) = 0, \qquad \Phi(a,a;x,y) = W(ax,ay). \tag{1}$$

Since $2 = 0$, the first identity gives the symmetry $\Phi(a,b;x,y) = \Phi(a,b;y,x)$. We will contradict the existence of such a Φ.

4. Step 2: a finite parity identity

Fix a finite block $I \subset \mathbb{N}$ of even size $2m$ and $a', b' \in E$ with $\|a'\|, \|b'\| \le 1$ (in Step 4 they will carry the earlier blocks). Let X_i, Y_i $(i \in I)$ be indeterminates. Expanding Φ multilinearly in $x = \sum_{i\in I} X_i e_i$ and $y = \sum_{i\in I} Y_i e_i$ defines polynomials

$$G_{pq} := \Phi(a' + x,\, b' + y;\, a' + x,\, b' + y)_{pq} \in k[X,Y] \qquad (p,q \in I),$$

whose values at points $x, y \in k^I$ (vectors supported on I) are the entries of an $I \times I$ matrix $G(x,y)$. Every element of F is a symmetric matrix with zero diagonal, since $W(x,y)_{pq} = W(x,y)_{qp}$ and $W(x,y)_{pp} = 2x_p y_p = 0$, and both properties pass to the closed span. So the matrix G is symmetric with zero diagonal. Each G_{pq} has degree ≤ 2 in X and ≤ 2 in Y, its part of bidegree $(2,2)$ is $g_{pq} := \Phi(x,y;x,y)_{pq}$, and $G_{pq} - g_{pq}$ has total degree ≤ 3.

Lemma 4. *For $p \ne q$ we have, in $k[X,Y]$,*

$$g_{pq}(Y,X) = g_{pq}(X,Y) + X_p^2 Y_q^2 + X_q^2 Y_p^2 \qquad \text{and} \qquad g_{pq}(X,X) = 0.$$

Proof. A field with a nontrivial absolute value is infinite, so it suffices to check these at points $x, y \in k^I$. By the symmetry in the last two slots, $g(y,x) = \Phi(y,x;y,x) = \Phi(y,x;x,y)$, so by bilinearity in the first two slots and (1)

$$\begin{aligned} g(x,y) + g(y,x) &= \Phi(x+y, x+y; x, y) + \Phi(x,x;x,y) + \Phi(y,y;x,y) \\ &= W\big((x+y)x, (x+y)y\big) + W(x^2, xy) + W(xy, y^2). \end{aligned}$$

Expanding, $W((x+y)x, (x+y)y) = W(x^2,xy) + W(x^2,y^2) + W(xy,xy) + W(xy,y^2)$. The first and last terms cancel against the two other summands and $W(xy,xy) = 0$, which leaves $W(x^2,y^2)$, whose (p,q) entry is $x_p^2 y_q^2 + x_q^2 y_p^2$. Finally $g(x,x) = W(x^2,x^2) = 0$. □

Let τ be the automorphism of $k[X,Y]$ swapping $X_i \leftrightarrow Y_i$, and δ the substitution $Y_i := X_i$.

Lemma 5 (balanced coefficients)**.** *Let $f \in k[X,Y]$ with $\tau f = f$ and $\operatorname{char} k = 2$. For every $S \subseteq I$ the coefficient of $\prod_{i\in S} X_i Y_i$ in f equals the coefficient of $\prod_{i\in S} X_i^2$ in δf.*

Proof. The coefficient of $\prod_S X_i^2$ in δf is the sum of the coefficients in f of all monomials $X^\alpha Y^\beta$ with $\alpha + \beta = 2 \cdot \mathbf{1}_S$. The involution $(\alpha,\beta) \mapsto (\beta,\alpha)$ of this set of monomials preserves coefficients (as $\tau f = f$), and its only fixed point is $\alpha = \beta = \mathbf{1}_S$; all other terms cancel in pairs. □

Proposition 6. *Let M be a perfect matching of I and $\nu := \prod_{i\in I} X_iY_i$. Then*

$$\operatorname{coeff}_\nu \prod_{\{p,q\}\in M} G_{pq} = 1.$$

Proof. The monomial ν has total degree $4m$, the maximum possible, so only the top parts contribute:

$$\operatorname{coeff}_\nu \prod_M G_{pq} = \operatorname{coeff}_\nu \prod_M g_{pq}.$$

Put $g'_{pq} := g_{pq} + X_p^2Y_q^2$. By Lemma 4,

$$\tau g'_{pq} = g'_{pq} \qquad \text{and} \qquad \delta g'_{pq} = X_p^2X_q^2.$$

A product containing a factor $X_p^2Y_q^2$ has no squarefree monomial, so replacing every g_{pq} by g'_{pq} does not change the coefficient of ν. By Lemma 5, applied to the τ-invariant product $\prod_M g'_{pq}$, this coefficient is the coefficient of $\prod_{i\in I} X_i^2$ in

$$\delta \prod_M g'_{pq} = \prod_{\{p,q\}\in M} X_p^2X_q^2 = \prod_{i\in I} X_i^2,$$

namely 1. □

Lemma 7. *Let A be a symmetric matrix with zero diagonal over a commutative ring of characteristic 2, of even size $2m$. Then $\det A = \sum_M \prod_{\{p,q\}\in M} A_{pq}^2$, the sum being over perfect matchings; and the number of perfect matchings, $(2m-1)!!$, is odd.*

Proof. In $\det A = \sum_\sigma \prod_i A_{i\sigma(i)}$ (all signs are 1), σ and σ^{-1} contribute equally by symmetry, so only involutions survive; those with a fixed point contribute 0; a fixed-point-free involution is a perfect matching and contributes $\prod A_{pq}^2$. □

Lemma 8 (hypercube sum)**.** *Let R be a commutative ring of characteristic 2 and let $f \in R[Z_1,\dots,Z_N]$ have total degree $\le N$. Then $\sum_{z\in\{0,1\}^N} f(z)$ is the coefficient of $Z_1Z_2\cdots Z_N$ in f.*

Proof. By linearity we may take f to be a monomial Z^α. At a point $z \in \{0,1\}^N$ it takes the value 1 if $z_i = 1$ for all i in the support of α, and 0 otherwise; so the sum is $2^{N-|\operatorname{supp}\alpha|}$, which vanishes in R unless every variable occurs in Z^α. The only monomial of degree $\le N$ in which all N variables occur is $Z_1\cdots Z_N$. □

Corollary 9. *The values of $\det G$ at the $0/1$ points sum to 1,*

$$\sum_{x,y\in\{0,1\}^I} \det G(x,y) = 1,$$

and consequently there are $x, y \in \{0,1\}^I$ with $|\det G(x,y)| \ge 1$.

Proof. For a perfect matching M the product $\prod_{\{p,q\}\in M} G_{pq}$ has total degree $\le 4m$, which is the number of variables X_i, Y_i. By Lemma 8 and Proposition 6 its sum over the $0/1$ points is therefore $\operatorname{coeff}_\nu \prod_M G_{pq} = 1$. Squaring is additive in characteristic 2, so by Lemma 7

$$\sum_{x,y\in\{0,1\}^I} \det G(x,y) = \sum_M \Big(\sum_{x,y\in\{0,1\}^I} \prod_{\{p,q\}\in M} G_{pq}(x,y) \Big)^2 = (2m-1)!! \cdot 1 = 1,$$

because $(2m-1)!!$ is odd. If all the values $\det G(x,y)$ had absolute value < 1, the ultrametric inequality would give $|1| < 1$. □

The point is that the coefficient 1 does not depend on the unknown lift P: all of its freedom is carried by τ-invariant terms vanishing on the diagonal $Y = X$, which Lemma 5 cannot see. No single pair (x, y) works for every lift, but the 2^{4m} vertices of the cube cannot all fail at once.

5. Step 3: locality on ℓ^∞

For $x \in \ell^\infty$ write $x|_{\geq N}$ for x with its coordinates below N set to 0.

Proposition 10. *Under* (K2)*, a bounded linear functional φ on ℓ^∞ with $\varphi(e_i) = 0$ for all i is zero.*

Proof. φ kills finitely supported sequences, so $|\varphi(x)| \leq \|\varphi\| \sup_{i \geq M} |x_i|$ for every M. Suppose $\varphi \neq 0$. For every $0 < s < \|\varphi\|$ there is w with $\varphi(w) = 1$ and $\|w\| \leq 1/s$; fix one such vector w_0. Because $r_{n+1} < r_n$ we can choose $s_n \in \ell^\infty$ with $\varphi(s_n) = c_{n+2} - c_{n+1}$ and $\|s_n\| \leq r_n/\|\varphi\|$: take $s_n = (c_{n+2} - c_{n+1})w$ with $\|w\| \leq r_n/(r_{n+1}\|\varphi\|)$. Let $x_n := c_1 w_0 + \sum_{k<n} s_k$; then $\varphi(x_n) = c_{n+1}$ and $\|x_{n+d} - x_n\| \leq r_n/\|\varphi\|$ by the ultrametric inequality. The diagonal sequence $y_i := x_i(i)$ is bounded, since $\|x_i\| \leq \max(\|x_0\|, r_0/\|\varphi\|)$, and $|y_i - x_n(i)| \leq r_n/\|\varphi\|$ for $i \geq n$. Hence $|\varphi(y) - c_{n+1}| = |\varphi(y - x_n)| \leq r_n$, and since $|c_{n+1} - c_n| \leq r_n$ the ultrametric inequality gives $|\varphi(y) - c_n| \leq r_n$ for every n. So $\varphi(y) \in \bigcap_n B(c_n, r_n)$, a contradiction. □

This is the direction of van der Put's theorem [6] that we need, $(\ell^\infty/c_0)' = 0$; the index shift absorbs the usual $(1 + \varepsilon)$ loss, and no Hahn–Banach argument is used.

Proposition 11 (multilinear Phillips property)**.** *Let λ be a bounded d-linear form on ℓ^∞ and $\varepsilon > 0$. There is N such that $|\lambda(u_1, \ldots, u_d)| \leq \varepsilon$ whenever $\|u_j\| \leq 1$ for all j and $u_j = u_j|_{\geq N}$ for some j.*

Sketch (the complete proof is in the Lean development). *The case $d = 1$, $\lambda = \varphi$.* (a) $\varphi(e_i) \to 0$. Otherwise $|\varphi(e_{i_k})| \geq \varepsilon$ along a subsequence. Let T be the bounded operator with $T(x)_{i_k} = x_k/\varphi(e_{i_k})$ and all other coordinates 0, and $\psi := \varphi \circ T$, so that $\psi(e_k) = 1$ for all k. With sh the right shift, $\psi \circ (1 - \mathrm{sh})$ kills every e_k, hence vanishes by Proposition 10; but $(1 - \mathrm{sh})\mathbf{1} = e_0$, so $0 = \psi(e_0) = 1$. (b) For fixed x, $\varphi(x|_{\geq N}) - \varphi(x|_{\geq N+1}) = x_N \varphi(e_N) \to 0$, so $\varphi(x|_{\geq N})$ converges; the limit is a bounded linear functional of x killing every e_i, hence it is 0. (c) If the claim failed for ε, then (b) would give unit vectors w_k with disjoint finite supports moving to infinity and $|\varphi(w_k)| > \varepsilon$; the functional $c \mapsto \varphi(\sum_k c_k w_k)$ (coordinatewise sum) then contradicts (a).

The case $d > 1$, by induction on d. Suppose the claim fails for ε. First, for every M there is a unit tuple with $|\lambda| > \varepsilon$ *all* of whose slots vanish below M: start from a witness one of whose slots is supported very far out, and split any other slot into its part below M, a finite combination of the e_i, and its tail; the inductive hypothesis, applied to the finitely many $(d-1)$-linear forms obtained by freezing that slot at e_i, $i < M$, shows that the part below M contributes at most ε, so the tail carries $|\lambda| > \varepsilon$. Next, by (b) each slot can be cut down to a finite window. Finally choose such tuples $z^{(s)}$ with windows moving to infinity fast enough and let x be the coordinatewise sum of their first slots. Then $\lambda(x, z_2^{(s)}, \ldots, z_d^{(s)})$ equals $\lambda(z^{(s)})$, plus finitely many terms with an earlier first slot (each $\leq \varepsilon$ by the inductive hypothesis for $\lambda(z_1^{(r)}, \cdot)$, $r < s$), plus one term with a far first slot ($\leq \varepsilon$ by the case $d = 1$). So it has absolute value $> \varepsilon$ for every s, contradicting the inductive hypothesis for the $(d-1)$-linear form $\lambda(x, \cdot)$. □

This is exactly where ℓ^∞ and (K2) are used; the statement is false for c_0 and for spherically complete fields.

6. Step 4: gluing

Choose inductively blocks $I_1 < I_2 < \cdots$ with $|I_t| = 2t$ and vectors $a_t, b_t \in \{0,1\}^{I_t}$ as follows. Given the earlier stages let $a' = \sum_{s<t} a_s$, $b' = \sum_{s<t} b_s$, and apply Corollary 9 to get a_t, b_t with $|\det G_t| \geq 1$, where $G_t = \Phi(a' + a_t, b' + b_t; a' + a_t, b' + b_t)|_{I_t \times I_t}$. Put $\varepsilon_t := \frac{1}{2} C^{-2t}$ and use Proposition 11 (with $d = 4$) for the finitely many entry forms $\Phi(\cdot)_{pq}$, $p, q \in I_t$, to place all later blocks so far out that these forms are at most ε_t on unit arguments one of which is supported there.

Let $a, b \in \ell^\infty$ be defined by $a|_{I_t} = a_t$, $b|_{I_t} = b_t$ and zero elsewhere. These are not norm-convergent sums; $a = \mathbf{1}_A$ and $b = \mathbf{1}_B$ are indicator functions of two subsets of $\mathbb{N}$, so m_a, m_b are coordinate projections. Let $\mathcal{O} := \Phi(a, b; a, b) \in F$. Splitting $a = a^{\leq t} + a^{>t}$, $b = b^{\leq t} + b^{>t}$ and using multilinearity, $\|\mathcal{O}|_{I_t \times I_t} - G_t\| \leq \varepsilon_t$. All entries are bounded by C. Expanding the determinant multilinearly in columns, every term containing a column of the perturbation is at most $\varepsilon_t C^{2t-1} < 1$, so the ultrametric inequality gives

$$|\det \mathcal{O}|_{I_t \times I_t}| = |\det G_t| \geq 1 \qquad \text{for every } t.$$

On the other hand, every element of F is a uniform limit of matrices of finite rank, because each $W(x, y)$ has rank ≤ 2. So $\mathcal{O} \in F$ is within $\eta := (2C)^{-1}$ of a matrix R of some finite rank r. Expanding $\det(R + (\mathcal{O} - R))$ on $I_t \times I_t$ in the same way, terms with more than r columns from R vanish, so $|\det \mathcal{O}|_{I_t \times I_t}| \leq C^r \eta^{2t-r} < 1$ once $t > r$. This contradiction proves Theorem 2. □

7. An explicit field

The purpose of this section is to construct a field satisfying (K1) and (K2). Towards that end, let $k_0 = \mathbb{F}_2$ (or any field of characteristic 2). For $j \geq 0$ let $k_0((t^{1/2^j}))$ be the field of *formal* Laurent series $\sum_{n \geq N} a_n t^{n/2^j}$ ($N \in \mathbb{Z}$, $a_n \in k_0$) in the single variable $t^{1/2^j}$. These fields form an increasing chain, and we let

$$\mathcal{P} := \bigcup_{j \geq 0} k_0((t^{1/2^j})), \qquad |f| := 2^{-\operatorname{ord} f},$$

be the field of **2-power Puiseux series**. Here $\operatorname{ord} f$ is the minimum q for which $\operatorname{coeff}_{t^q} f \neq 0$. We set $|0| = 0$ by definition. Let k be the completion of $\mathcal{P}$ for this (ultrametric) absolute value. This is characteristic 2 since $k_0 \hookrightarrow \mathcal{P} \hookrightarrow k$.

Proposition 12. *k is not spherically complete.*

Proof. We exhibit a sequence of balls as in (K2). Put

$$\gamma_m := 1 - 2^{-(m+1)}, \qquad c_n := \sum_{m=0}^{n} t^{\gamma_m} \in \mathcal{P}, \qquad r_n := 2^{-\gamma_{n+1}}.$$

The balls have the form required in (K2). Since $c_{n+1} - c_n = t^{\gamma_{n+1}}$ we have $|c_{n+1} - c_n| = 2^{-\gamma_{n+1}} = r_n$. Since the γ_m are strictly increasing to 1, the r_n are strictly decreasing, and $r_n > 2^{-1} = \frac{1}{2}$.

The intersection is empty. Suppose that $a \in k$ satisfies $|a - c_n| \leq r_n$ for all n. As $\mathcal{P}$ is dense in k there is $y \in \mathcal{P}$ with $|a - y| < \frac{1}{2}$, and then for every n

$$|y - c_n| \leq \max\big(|y - a|, |a - c_n|\big) \leq \max\big(\tfrac{1}{2}, r_n\big) = r_n. \tag{2}$$

Now $y \in k_0((t^{1/2^j}))$ for some j. The exponent $\gamma_j = (2^{j+1} - 1)/2^{j+1}$ has an odd numerator, therefore γ_j is not a multiple of $1/2^j$ and so $\operatorname{coeff}_{t^{\gamma_j}} y = 0$. For $n \geq j$ we have $2^{-\gamma_j} > 2^{-\gamma_{n+1}} =$

r_n. Furthermore, the term t^{γ_j} occurs in c_n, so

$$\operatorname{coeff}_{t^{\gamma_j}}(y - c_n) = 0 - 1 \neq 0, \qquad \text{hence} \qquad \operatorname{ord}(y - c_n) \leq \gamma_j \quad \text{and} \quad |y - c_n| \geq 2^{-\gamma_j} > r_n,$$

which contradicts (2). □

8. Remarks

Remark 13 (Why characteristic 2, why ℓ^∞, why this F). In characteristic $\neq 2$ the lift $\frac{1}{2}\big(m \circ (f,g) + m \circ (g,f)\big)$ exists, and the parity argument breaks (pairs add instead of cancelling). If E has an orthogonal base (e_i), e.g. $E = c_0$, and F is Banach, then

$$P(f,g)(m)(x,y) = \sum_{i<j} (x_i y_j - x_j y_i)\, m(f e_i, g e_j)$$

is a bounded lift in every characteristic, so Q is analytic; a similar formula works for a base of E', and t-orthogonal bases suffice. This covers finite-dimensional E and all Banach spaces over a discretely valued k [7], so the counterexample needs a dense valuation; one can also show that Q is analytic when F is spherically complete. On c_0 the glued vectors a, b of Step 4 do not exist and Proposition 11 fails. For $E = \ell^\infty$ and the larger target $\ell^\infty(\mathbb{N} \times \mathbb{N}, k) \supseteq F$ a bounded lift also exists (sort the indices entrywise), but it sends W to matrices such as the "staircase" $\mathbf{1}_{\{(p,q)\,:\,p \text{ odd}, q \text{ even}, p<q\}}$, which are not limits of finite-rank matrices, so it leaves F. These positive statements are not proved here and are not part of the formalization.

Remark 14 (What is forced). No single pair (x, y) works for every lift: adding to Φ a bounded 4-linear map with values in F that is alternating in (a, b) and in (x, y) gives another admissible Φ, and this can change a given off-diagonal entry of $G(x, y)$ by an arbitrary scalar. Blocks of bounded size cannot give a contradiction either: for $u = \mathbf{1}_{\text{odd}}$, $v = \mathbf{1}_{\text{even}}$ the rank-two matrix $W(u, v) \in F$ has the entry 1 in every diagonal block $\{2s - 1, 2s\}$. On a finite block the sorted lift of the previous remark has norm 1 and satisfies all the identities of Step 2, so there is no finite-dimensional obstruction: the invariant is the rank of one element of F against the size of the blocks, and Step 3 is what allows infinitely many blocks to be glued into one element.

Remark 15 (Higher degree, other characteristics). On paper, in characteristic 2 every $|\iota| \geq 2$ is covered by wedging with a coordinate of $E \oplus k$; this is not formalized. In odd characteristic ℓ, for forms of degree ℓ, the question is open: odd-degree forms have no Pfaffian-type top power, so a different invariant is needed.

9. Formalization

The Lean 4/Mathlib development is at

https://github.com/Deicyde/alternating-analytic-counterexample

The library `AltCounter/` contains no `sorry`, and the only axioms it uses are the three standard ones (`propext`, `Quot.sound` and `Classical.choice`). Its headline theorem assumes that Mathlib's

`ContinuousAlternatingMap.compContinuousLinearMapCLM`

is C^ω for all nontrivially normed fields, normed spaces and finite index types, and derives `False`; a second theorem does the same assuming this only for complete ultrametric fields of characteristic 2, Banach spaces, $E' = E$ and $|\iota| = 2$. Both are restated in a Mathlib-only file `Challenge.lean` (with `sorry` in place of the proofs, as that format requires) and checked against `Solution.lean` by the `leanprover/comparator` tool. Steps 1–4 and §7 are formalized, including the complete proof of Proposition 11 by induction on d; Lemma 8 is `sum_eval_cubePt`, and the field of §7 is built with $k_0 = \mathbb{F}_2$. The Lean constants in Step 4

are chosen with more slack than here, and the Lean form of Corollary 9 records only that the witnesses lie in the unit ball, which is all that Step 4 uses. Not formalized: non-analyticity at base points other than 0, index types with more than two elements, the positive statements in the remarks, and the bundle-level consequence (no C^ω vector-bundle structure on bundles of continuous alternating maps in this generality).

References

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